

From Macro Goals to Adaptive Skills: Dialogue Policy Learning under Intent Drift

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Abstract—Traditional goal-oriented dialog agents require inefficient retraining when new objectives arise. While recent multi-objective methods parameterize goals, they rely on rigid numerical preference vectors. This misaligns with real-world settings, where macro-goals are expressed in natural language, requiring strategic flexibility and non-negotiable constraints. Furthermore, existing policies ignore intent drift, maintaining static behavior rather than adapting to evolving dialog contexts. To address these gaps, we propose ALMOND, a novel hierarchical framework for learning dialog policies. A *low-level policy* executes communication skills via a regret-based mechanism, prioritizing unmastered skills to bootstrap complex behaviors robustly. Concurrently, a *high-level policy* adaptively selects short-term skills by monitoring intent shifts, leveraging strategic hints distilled during training to satisfy the overarching linguistic goal. Extensive experiments show ALMOND significantly outperforms baselines. Our framework achieves superior linguistic goal completion, minimizes constraint violations, drastically reduces adaptation latency, and establishes a new state-of-the-art for short-term success rates.

Index Terms—goal-oriented dialogue, multi-objective reinforcement learning, intent drift, dialogue policy planning, negotiation, conversational recommendation

I. INTRODUCTION

Goal-oriented dialogue agents are increasingly expected to act on behalf of users, not simply to generate fluent responses. In buyer-side negotiation, for example, a useful agent must pursue a favorable price while keeping the seller engaged, reaching an agreement within a limited number of turns, and respecting the buyer’s budget. Target-driven recommendation involves a similar balancing act: the agent should guide the user toward a desired item without degrading satisfaction or conversational efficiency. These objectives are not independent. An aggressive bargaining move may improve price utility if the seller remains cooperative, yet the same move can reduce the probability of closing the deal. The policy therefore has to reason about strategic trade-offs and constraints as part of the dialogue process itself.

The difficulty is compounded by the way goals are specified in practice. Traditional goal-oriented policies are usually trained for a fixed objective, which makes adaptation to new user requirements expensive. Recent multi-objective methods reduce this cost by conditioning a policy on preference vectors, but the interface remains rigid: it assumes that the user’s goal has already been translated into explicit numerical weights. In realistic deployment, users are more likely to provide natural-language macro-goals, such as “buy cheaply, but do not lose the seller if the item seems scarce.” Such instructions combine soft preferences, situational conditions, and hard constraints. Their meaning is also local. Early in a negotiation, the agent may reasonably emphasize price gain; after the seller becomes firm, issues a final offer, or signals walkaway risk, the same macro-goal may require a temporary shift toward deal probability, rapport preservation, or turn efficiency. A single static vector can describe one trade-off, but it does not explain how the agent should change its short-term behavior as the conversation evolves.

Prior work on target-guided and proactive dialogue has made substantial progress on multi-turn planning. Target-guided models learn keyword or topic transitions toward a predefined target [1]–[5], while recent planners use Brownian-bridge trajectories, LLM reasoning, self-reflection, experience retrieval, and dual-process planning to improve future-oriented behavior [6]–[10]. This literature shows that dialogue systems benefit from planning beyond the next utterance. At the same time, much of it assumes that the target is fixed before the interaction is optimized. The setting we study is different: the macro-goal remains stable at the linguistic level, but the short-term skill needed to pursue it can change as the other party’s intent and stance change.

The closest multi-objective predecessor to our work is PADPP, which formulates goal-oriented dialogue as preference-conditioned policy learning and trains a single planner that adapts to different objective vectors at inference

time [11]. Its use of generalized policy improvement (GPI) connects naturally to the broader line of work on reusable behavior across reward functions [12], [13]. PADPP substantially reduces the need to retrain a separate policy for every objective setting. However, it still assumes that the preference vector is given. This leaves open a practical question at the interface between human intent and policy control: how should an ambiguous macro-goal be decomposed into short-term dialogue skills, when should the active skill change, and how should that change be justified by evidence in the conversation?

This question becomes especially important under conversational intent drift. Fine-grained intention understanding suggests that dialogue intent involves multiple aspects, including situation, emotion, action, and knowledge [14], and recent drift detection work studies how active intent changes over continuous conversations [15]. In our buyer-side negotiation setting, drift appears as a change in the seller’s stance: a seller may begin neutral, become firm after repeated low offers, issue a final offer, or signal walkaway risk. If the buyer continues executing the same short-term strategy, it may bargain past the point at which the seller’s stance has changed; if it switches too early, it may sacrifice price utility unnecessarily. We therefore study goal-driven dialogue policy learning in which a stable natural-language macro-goal must be served through adaptive short-term skills under intent drift and hard constraints.

We propose **ALMOND**, a hierarchical framework for learning adaptive dialogue policies from natural-language macro-goals. ALMOND separates the problem into two coupled levels. At the low level, the policy learns reusable communication skills, such as bargaining, conceding, asking for clarification, or preserving rapport. Because complex dialogue behaviors often depend on skills that are unevenly learned, ALMOND uses a regret-based mechanism to give more training attention to under-mastered skills rather than assuming that all skills progress uniformly. At the high level, a policy monitors the dialogue state for intent shifts and selects short-term skills conditioned on the macro-goal, the recent interaction history, and strategic hints distilled during training. In this design, language-level goal interpretation guides strategic skill selection, while the learned low-level policy remains responsible for executing concrete communicative behavior.

We evaluate ALMOND with metrics that reflect constrained adaptation rather than final task completion alone. LLM-SR measures linguistic goal completion using an LLM-as-Judge protocol with human-verification samples. Goal Success Rate (GSR) counts a dialogue as successful only when the buyer reaches a deal within the buyer’s budget ceiling and turn limit. Turn-to-Drift Adaptation Delay (T2DA) measures $t_{\text{adapt}} - t_{\text{drift}}$, and Constraint Violation Rate (CVR) tracks blocked and executed safety violations. Across our benchmark settings, ALMOND improves LLM-as-Judge goal completion by 4–7 percentage points and reduces adaptation delay by 2–3 turns over static-weight and ablation baselines. Phase-1 skill training further improves r -fair, r -gain, and r -deal by 3–6 percentage points while reducing average turns.

Our contributions are:

- We formulate goal-driven dialogue from ambiguous natural-language macro-goals under conversational intent drift and hard constraints.
- We propose ALMOND, a hierarchical framework that combines regret-aware low-level skill learning with drift-aware high-level skill selection.
- We use strategic hints distilled during training to guide short-term skill selection toward the overarching linguistic goal.
- We construct persona-conditioned seller simulators with deterministic drift triggers and evaluate constrained adaptation with LLM-SR, GSR, T2DA, and CVR.

II. RELATED WORK

a) Target-guided and proactive dialogue planning.

Target-guided dialogue studies how an agent can steer a conversation toward a specified topic, target, or endpoint. Early methods model this process through keyword prediction, discourse constraints, commonsense knowledge, and global planning over knowledge graphs [1]–[5]. More recent proactive planners broaden the planning signal with stochastic path modeling, action-topic planning, LLM-based reasoning, reflection, and consistency correction [6]–[8], [16], [17]. These approaches are closely related in spirit because they treat dialogue as a long-horizon control problem rather than a sequence of independent responses. Their targets, however, are usually specified before planning. ALMOND considers a complementary setting in which the macro-goal remains expressed in language, but the short-term skill needed to pursue it changes with the evolving dialogue state.

b) Recommendation and negotiation foundations. Conversational recommendation provides data and tasks for proactive target pursuit. DuRecDial and DuRecDial 2.0 introduce multi-domain recommendation dialogues [18], [19], while target-driven recommendation methods plan action-topic paths or retrieve past experience to anticipate future states [9], [20]. Negotiation dialogue research studies a different but equally strategic setting, where agents learn to bargain through end-to-end optimization, self-play, or decoupled strategy and language generation [21], [22]. These domains motivate the skill space used in ALMOND, including bargaining, concession, rapport maintenance, clarification, and deal-seeking behavior. They also motivate the evaluation protocol: a dialogue may reach an acceptable final outcome while still violating a budget constraint, taking too many turns, or failing to adapt when the counterparty’s intent shifts.

c) Adaptive dialogue policy learning. A growing line of work separates strategic dialogue planning from surface-form generation. PPDPP trains a plug-and-play policy planner with goal-oriented feedback [23], and TRIP improves planning in non-collaborative dialogue through diversified user simulation [24]. PADPP is the closest multi-objective predecessor: it learns a planner that can adapt to different numeric preference vectors at inference time [11]. This makes PADPP a strong foundation for studying preference-adaptive dialogue,

but it also clarifies the interface that remains unmodeled. In ALMOND, the control signal is not an observed vector; it is a short-term skill selected under a natural-language macro-goal and revised as the conversation changes. This shift is motivated by fine-grained intention understanding and intent-drift detection [14], [15], which emphasize that dialogue state is not only informational but also strategic and affective.

d) Dynamic preferences and constrained skill learning.: Outside dialogue, dynamic-weight MORL studies policies whose objective weights change over time [25], while generalized MORL learns policies over a preference space [26]. Surveys of MORL clarify the trade-offs among utility functions, Pareto coverage, and planning assumptions [27]. Successor features and GPI provide a mechanism for reusing value knowledge across reward functions [12], [13], and constrained policy optimization motivates treating safety requirements as constraints rather than only as shaped rewards [28]. ALMOND draws on this view of reusable behavior under changing objectives, but moves the adaptation signal closer to dialogue practice: short-term skills are inferred from a linguistic macro-goal, recent interaction history, and drift evidence. The regret-based low-level mechanism further biases training toward skills that remain under-mastered, allowing more complex communicative behavior to be bootstrapped without retraining a new policy for every objective.

III. PROBLEM FORMULATION

Following prior studies [10], we formulate the goal-oriented dialogue process as an NL-POMDP (Natural Language-conditioned Partially Observable Markov Decision Process). Formally, the system is provided with a conversational macro-objective g , expressed in natural language, which it must satisfy by the conclusion of the dialogue.

At each dialogue turn t , the system observes the current state s_t , which is defined by the dialogue history:

$$s_t = \{u_1^{sys}, u_1^{usr}, \dots, u_t^{sys}, u_t^{usr}\} \quad (1)$$

where u^{sys} and u^{usr} denote the utterances generated by the system and the user, respectively. Based on s_t and the overarching goal g , the dialogue agent selects a strategic action $a_t \in \mathcal{A}$, where $\mathcal{A}$ represents a pre-defined set of candidate dialogue strategies formulated by domain experts. The system's behavior is governed by a dialog policy π_θ , parameterized by θ , which serves as a mapping function from the current state and natural language goal to a probability distribution over the action space:

$$a_t \sim \pi_\theta(a|s_t, g) \quad (2)$$

Guided by the selected action a_t , the system player generates the subsequent utterance u_{t+1}^{sys} . In return, the user player produces a response u_{t+1}^{usr} . Crucially, the user's response is governed by their underlying intent $i_t \in \mathcal{I}$. Under the POMDP formulation, this intent i_t acts as a hidden state that is strictly unobservable to the system. The agent only perceives the emitted textual observation u_{t+1}^{usr} , rendering the true intent a latent variable. In real-world negotiations, user intent is

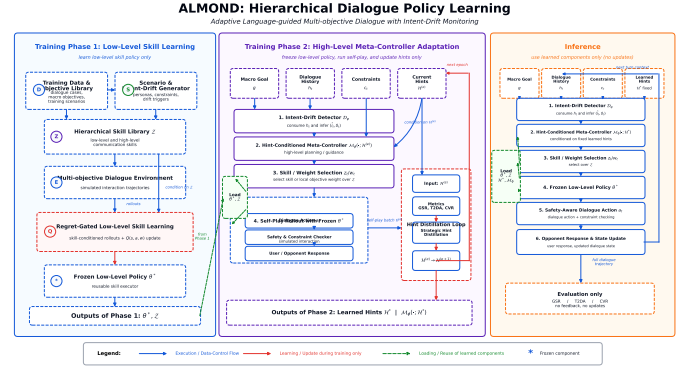


Fig. 1. ALMOND decouples regret-based low-level skill learning from intent-aware high-level skill selection.

rarely static; thus, we explicitly model *intent drift* ($i_t \rightarrow i_{t+1}$) as the conversation unfolds. The user's response generation is formally modeled as being conditioned on the updated dialogue context (incorporating the system's latest response) and their dynamically drifting intent:

$$u_{t+1}^{usr} \sim P_{usr}(u|s_t \cup \{u_{t+1}^{sys}\}, i_{t+1}) \quad (3)$$

To steer the dialogue toward a successful resolution, the agent must not only pursue the macro-objective g but also continuously infer the unobservable intent from the dialogue history and adapt its actions to these latent intent shifts i_{t+1} .

This interactive process iterates until the macro-objective is successfully achieved or the dialogue reaches a maximum turn limit T . Let $\tau = (s_1, a_1, s_2, \dots, s_T)$ denote the complete dialogue trajectory. The ultimate objective is to learn an optimal dialog policy π_θ that maximizes the expected success rate. Let $\Phi(\tau, g) \in \{0, 1\}$ be a binary indicator function denoting whether the trajectory τ successfully fulfills the natural language macro-objective g . The objective function to be maximized is defined as:

$$\max_{\theta} \mathcal{J}(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [\Phi(\tau, g)] \quad (4)$$

IV. PROPOSED METHOD

ALMOND is designed for the setting where the agent receives a stable natural-language macro-goal, but the correct short-term behavior changes as the counterparty's intent drifts. The framework therefore separates reusable skill execution from macro-goal interpretation. The first phase trains a universal low-level policy over a continuous objective simplex. The second phase freezes this low-level policy and learns a high-level controller that maps the macro-goal and the current dialogue evidence into a local skill vector. Fig. 1 summarizes this two-phase control flow.

A. Phase 1: Regret-Based Universal Skill Learning

The low-level policy operates in the same multi-objective action space as preference-conditioned dialogue planners. In negotiation, a local skill is a weight vector $w \in \Delta^2$ over three primitive objectives: `sl_ratio`, `fairness`, and

deal_rate. These dimensions correspond respectively to buyer price gain, relationship-preserving fairness, and the probability of closing an agreement. Given a dialogue state s_t , action a_t , and local skill w , the low-level network predicts a vector action value $Q(s_t, a_t, w; \theta) \in \mathbb{R}^3$. The scalar reward used for a particular skill is

$$r_w(s_t, a_t) = w^\top r_t, \quad (5)$$

where $r_t = [r_t^{gain}, r_t^{fair}, r_t^{deal}]$ is the environment reward vector.

Anchor skill curriculum. Training directly on arbitrary Dirichlet samples can make the policy unstable, because early GPI targets may be produced by poorly learned preferences. ALMOND first anchors the low-level policy at a small set of reliable skills:

$$\mathcal{W}_0 = \{e_1, e_2, e_3, u, m_{12}, m_{23}, m_{13}\}. \quad (6)$$

Here e_i are the three one-hot corner weights, $u = [1/3, 1/3, 1/3]$, and m_{ij} are the three edge midpoints. Each Phase-1 episode samples one anchor and trains the policy with a self TD target, without a GPI teacher:

$$a_{\text{self}} = \arg \max_{a'} w^\top Q(s_{t+1}, a', w; \bar{\theta}), \quad (7)$$

$$y_{\text{self}} = w^\top (r_t + \gamma(1 - d_t)Q(s_{t+1}, a_{\text{self}}, w; \bar{\theta})), \quad (8)$$

$$\mathcal{L}_{\text{anchor}}(\theta) = \mathbb{E} \left[(y_{\text{self}} - w^\top Q(s_t, a_t, w; \theta))^2 \right]. \quad (9)$$

This curriculum produces a set of stable basic skills that initialize the converged-skill memory for later knowledge reuse.

Regret-gated knowledge reuse. After anchor training, ALMOND expands from discrete skills to the continuous simplex using a regret-aware variant of PADPP. We maintain a delayed snapshot $Q(\cdot; \theta^-)$ of the current network and estimate the regret of a candidate skill w on a replay-buffer state batch:

$$\text{Reg}(w) = \mathbb{E}_{s,a} [|Q(s, a, w; \theta) - Q(s, a, w; \theta^-)|]. \quad (10)$$

Only skills whose regret is below a threshold ϵ are admitted to the teacher set:

$$\mathcal{W}_{\text{conv}} = \{w \mid \text{Reg}(w) < \epsilon\}. \quad (11)$$

This gate prevents unstable or recently changed preferences from serving as teachers. For a sampled training skill w , the GPI teacher action is selected from converged skills only:

$$a_{\text{gpi}} = \arg \max_{a'} \max_{w_i \in \mathcal{W}_{\text{conv}}} w^\top Q(s_{t+1}, a', w_i; \bar{\theta}). \quad (12)$$

The knowledge-reuse target evaluates the teacher action under the current training skill:

$$y_{\text{know}} = r_t + \gamma(1 - d_t)Q(s_{t+1}, a_{\text{gpi}}, w; \bar{\theta}). \quad (13)$$

The final low-level update mixes independent execution with regret-gated GPI:

$$\mathcal{L}_{\text{low}}(\theta) = (1 - \alpha)\mathcal{L}_{\text{self}}(\theta) + \alpha\mathcal{L}_{\text{know}}(\theta). \quad (14)$$

Here $\mathcal{L}_{\text{self}}$ is the scalar DDQN-style TD loss under w , and $\mathcal{L}_{\text{know}}$ is the vector TD loss induced by a_{gpi} . During rollout, candidate skills can also be sampled in proportion to $\text{Reg}(w)^\beta$, which concentrates data collection on under-mastered regions of the skill simplex.

B. Phase 2: Intent-Aware High-Level Meta-Control

Phase 1 learns how to execute a local skill, but it does not decide which skill should be active for a natural-language macro-goal. In Phase 2, the low-level checkpoint θ^* is frozen. A high-level controller produces a local weight vector w_t from the macro-goal g , the dialogue state s_t , hard constraints c , the previous local skill, and the current estimate of seller intent. The frozen low-level policy then selects the action:

$$a_t = \arg \max_{a \in \mathcal{A}} w_t^\top Q(s_t, a, w_t; \theta^*). \quad (15)$$

Intent-drift detector. Because the true intent i_t is hidden, ALMOND uses a detector to infer intent from visible dialogue evidence:

$$(\delta_t, \hat{i}_t) = \mathcal{D}_\psi(s_t, \hat{i}_{t-1}, \mathcal{H}_{\text{det}}), \quad (16)$$

where δ_t indicates whether the seller's intent has changed. The intent catalog used in the implementation contains four states: *neutral*, *firm*, *final_offer*, and *walkaway_risk*. These states correspond to observable signals such as ordinary bargaining, hardened price resistance, take-it-or-leave-it offers, and frustrated threats to sell elsewhere.

High-policy weight selection. The high policy regenerates w_t at the beginning of the dialogue, when drift is detected, or after a fixed macro horizon K . Let

$$m_t = \mathbb{I}[t = 0 \vee \delta_t = 1 \vee t \bmod K = 0]. \quad (17)$$

The local skill is updated as

$$w_t = \begin{cases} \mathcal{M}_\phi(g, s_t, \hat{i}_t, c, w_{t-1}, \mathcal{H}_{\text{pol}}), & m_t = 1, \\ w_{t-1}, & m_t = 0. \end{cases} \quad (18)$$

In the LLM controller path, $\mathcal{M}_\phi$ first emits an inspectable natural-language allocation, e.g., percentages over price saving, fairness, and deal closing, and the allocation is parsed into a normalized vector $w_t = [w_t^{gain}, w_t^{fair}, w_t^{deal}]$. This explicit allocation makes the macro decision auditable: under the same macro-goal, the controller may push price gain while the seller is neutral but raise deal_rate and fairness when the seller becomes firm or signals walkaway risk.

Safety masking. ALMOND treats hard buyer constraints as action-level constraints rather than soft preferences. After the low-level policy proposes an action, a safety mask checks whether the committed price violates the buyer ceiling. If it does, the action is replaced by a ceiling-respecting counteroffer:

$$\tilde{a}_t = \mathcal{M}_{\text{safef}}(a_t, c). \quad (19)$$

This mask is deliberately outside the reward scalarization so that no temporary skill allocation can justify overpaying.

Hint distillation. The high-level controller is improved without retraining the low policy. During self-play, each

episode records the selected weights, detected intents, drift events, blocked violations, and final constrained metrics. A distiller turns these records into reusable natural-language hints:

$$\mathcal{H}^{(e+1)} = \text{Distill}_\omega \left(\mathcal{H}^{(e)}, \mathcal{B}^{(e)}, \text{GSR}^{(e)}, \text{T2DA}^{(e)}, \text{CVR}^{(e)} \right), \quad (20)$$

where $\mathcal{B}^{(e)}$ is the compact self-play digest at epoch e . In the two-agent variant, ALMOND keeps separate hint memories for the detector and the high policy. During training, gold simulator intent can supervise high-policy refreshes while detector predictions are scored; at inference time, the detector prediction drives the weight refresh. This design lets ALMOND adapt macro-goals through explicit, reusable skill vectors while preserving the general low-level competence learned in Phase 1.

V. EXPERIMENTAL

A. Experimental Setup

We design the experiments around three research questions.

- **RQ1: Low-level skill learning.** Does the proposed Regret-Based Universal Skill Learning module improve low-level policy learning? We compare the Phase-1/Phase-2 low policy against PADPP under the PADPP negotiation protocol, with the uniform weight setting as the primary comparison and the three corner objectives as diagnostic settings.
- **RQ2: End-to-end macro-goal adaptation.** Does the complete two-phase ALMOND model improve performance on Macro Goals, where the input objective is expressed in natural language and seller intent may drift during the dialogue?
- **RQ3: Ablation study.** Which components are responsible for the gains? We evaluate variants that remove regret-gated GPI, dynamic high-level weight selection, intent detection, hint distillation, and the safety mask.

Baselines. We compare ALMOND with four policy-planning baselines. *PADPP* [11] is the closest multi-objective baseline: it trains one preference-conditioned planner and uses a fixed numerical preference vector during an episode. In our macro-goal benchmark, the PADPP-static baseline uses the scenario’s initial mapped weight `static_w` throughout the dialogue. *PPDPP* [23] trains a plug-and-play policy planner from human data and goal-oriented AI feedback, but does not explicitly learn a dynamic local objective vector under intent drift. *DPDP* [10] combines a fast policy model with deliberative MCTS planning, providing a strong look-ahead planning baseline. *TRIP* [24] improves strategy planning for non-collaborative dialogue through user-aware planning and diversified user simulation. For RQ1, PADPP is the main baseline because both methods share the same low-level multi-objective interface $Q(s, a, w)$.

Datasets and scenario construction. We evaluate on two benchmark families: Bargain and Recommendation. Bargain is derived from CraigslistBargain negotiation cases [22]. For each case, we keep the item name, buyer and seller reference

prices, item descriptions, and source dialogue length. Recommendation is derived from the English DuRecDial 2.0 corpus [19]. Since DuRecDial is a recommendation corpus and does not contain transaction prices, we convert its movie, music, and point-of-interest cases into recommendation-derived negotiation scenarios with deterministic synthetic price bands generated from a fixed seed. This keeps the recommendation context reproducible while preserving the same buyer-seller evaluation interface.

The generated scenarios are balanced across four drift modes: *static_no_drift*, *gradual_firming*, *abrupt_final_offer*, and *frustrated_walkaway*. Each scenario contains a natural-language `macro_goal`, a buyer intent identifier, an initial static weight, a price ceiling, a target price, a turn limit, a seller persona, a deterministic drift trigger, and an expected direction of weight adaptation. Phase-1 experiments use the original negotiation policy-learning protocol with the seven anchor weights in $\mathcal{W}_0$ and evaluate static skills using the PADPP Table-2 style setting. Phase-2 experiments use the generated macro-goal scenarios: training runs self-play and hint distillation on the train splits, while final evaluation is conducted on the held-out test splits.

Evaluation metrics. For low-level policy learning (RQ1), we report the standard negotiation metrics used by PADPP: Success Rate (SR), average dialogue turns, r_{gain} , r_{fair} , and r_{deal} . Following the implementation, r_{gain} , r_{fair} , and r_{deal} are averaged over the agent decision turns in an episode.

For macro-goal adaptation (RQ2 and RQ3), we report four constrained adaptation metrics. *LLM-SR* is the judge-based rate at which the dialogue reaches the intended outcome, with human audit samples exported for verification. *Goal Success Rate* (GSR) is stricter:

$$\text{GSR} = \mathbb{I}[\text{deal} = 1 \wedge p_{\text{deal}} \leq p_{\text{ceiling}} \wedge T_{\text{dialogue}} \leq T_{\text{limit}}]. \quad (21)$$

Turn-to-Drift Adaptation Delay (T2DA) measures how quickly the controller changes its local skill after the seller intent drifts:

$$\text{T2DA} = t_{\text{adapt}} - t_{\text{drift}}, \quad (22)$$

where t_{adapt} is the first turn after drift where $\|w_t - w_{\text{pre}}\|_1 \geq 0.25$ and the shift follows the expected direction. *Constraint Violation Rate* (CVR) measures the fraction of turns that attempt or execute an over-ceiling buyer action; we report blocked and actual violations separately. Higher is better for SR, r_{gain} , r_{fair} , r_{deal} , LLM-SR, and GSR; lower is better for average turns, T2DA, and CVR.

B. Experimental Results

We organize the results according to the three research questions. The first block reports whether regret-based low-level training improves the static uniform and anchor-weight skills over PADPP. The second block evaluates the full two-phase ALMOND system on Bargain and Recommendation macro-goal scenarios. The third block reports ablations over regret gating, dynamic high-level control, intent detection, hint distillation, and safety masking.

TABLE I
GENERATED BENCHMARK SPLITS USED FOR ALMOND EVALUATION.

Benchmark	Train	Test	Construction
Bargain	1000	250	CraigslistBargain cases with item context, buyer/seller prices, macro-goals, buyer constraints, seller personas, deterministic drift modes, and expected post-drift weight shifts.
Recommendation	1000	250	DuRecDial-derived movie, music, and point-of-interest cases converted into the same buyer-seller interface with deterministic synthetic prices and recommendation-specific item context.

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